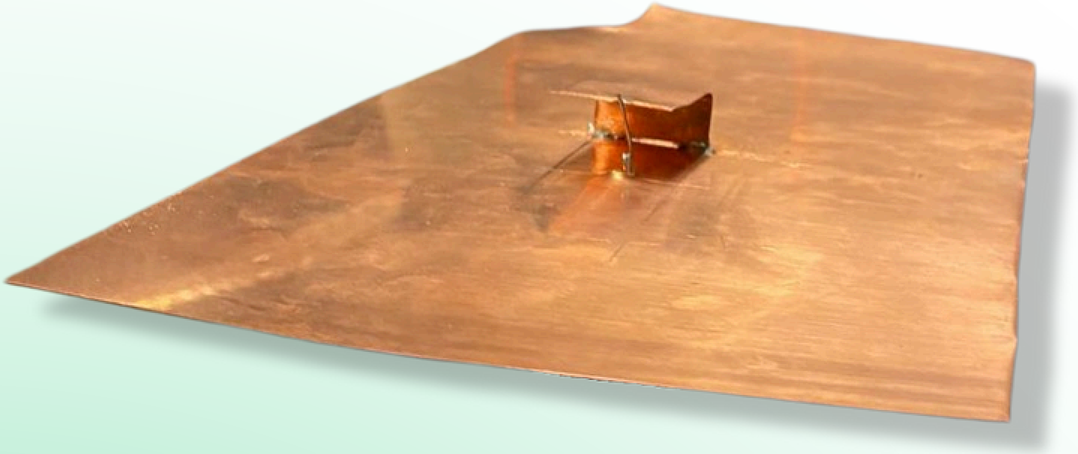

2.4GHz PIFA Antenna Datasheet

A compact, high-efficiency PIFA antenna designed for 2.4GHz WiFi applications, offering low-profile integration and optimized radiation for reliable wireless performance.

Presented by

José Pérez Clar
Ricardo Pereira González

Presented to

Radiocommunication Systems - 2025

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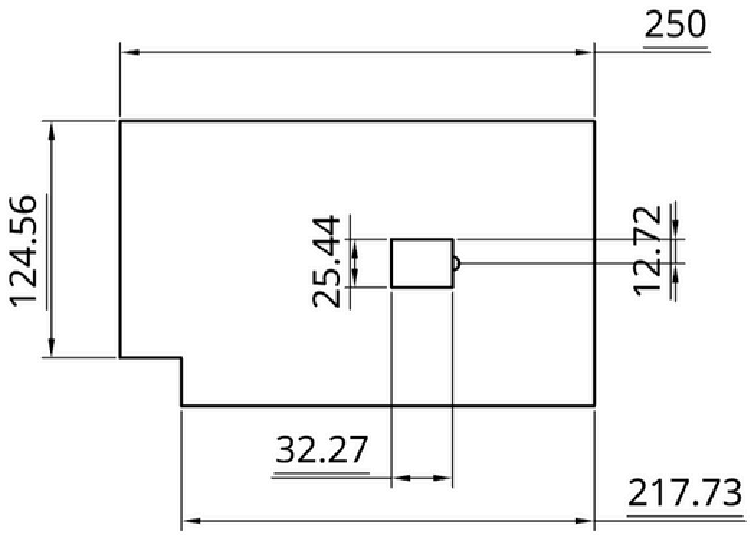
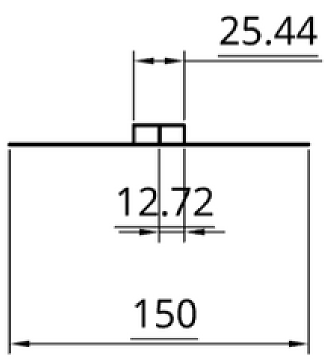
Dimensions of the antenna

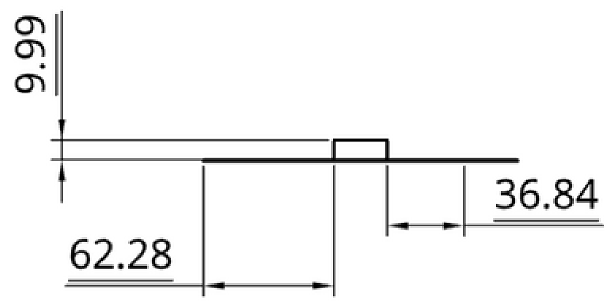
TOP



RIGHT



LEFT



Important Considerations

- All the distances are stated in millimetres (mm).
- The drawings above are not scaled.
- The connector hole has a diameter of 7 mm.
- The ground plane is approximated to 250x150mm.
- All values concerning the feed point and radiating patch are exact.

Behavioural Details

Frequency Band: 1.9 GHz to 2.65 GHz (at -6 dB)

Adaptation Values and Graphs

Return Loss:

- Minimum Return Loss: -9.27 dB
- Maximum Return Loss: 0.00 dB
- Return Loss at 2.4GHz: -7.89 dB

Impedance at 2.4GHz:

- Real Part (R): 51.0787 Ω
- Imaginary Part (X): 0.9298 Ω

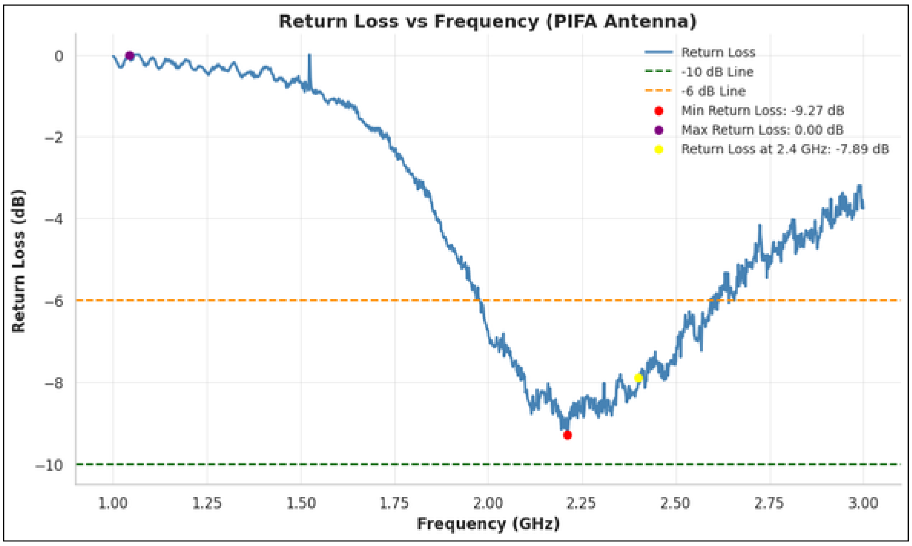


Figure 1: Return Loss graph

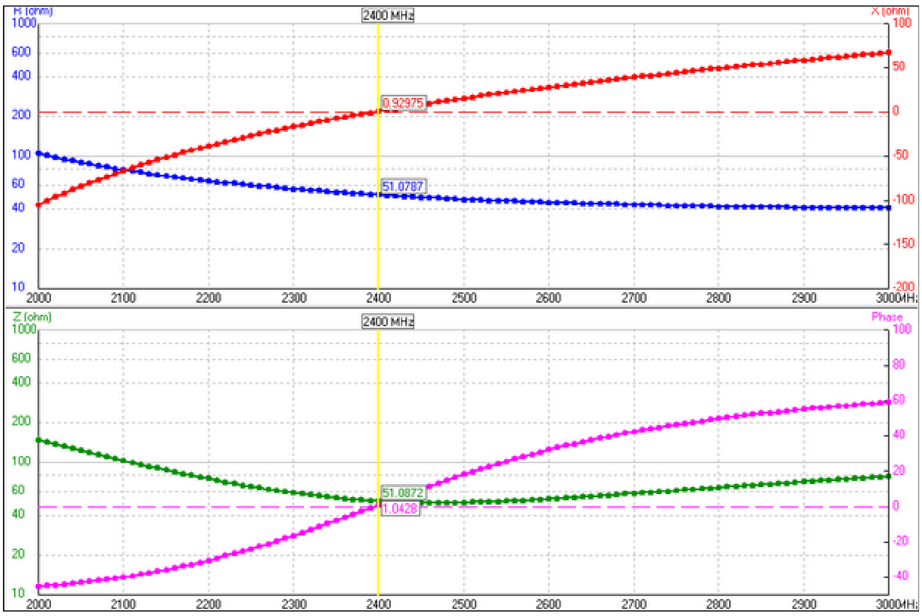


Figure 2: Impedances graph

Behavioural Details

Radiation Data at 2.4GHz

Max Gain: 5.33 dBi

Front-to-Back Ratio (F/B): 0.46 dB

Beamwidth:

- Vertical Plane (Elevation): 55°
- Horizontal Plane (Azimuth): 74°

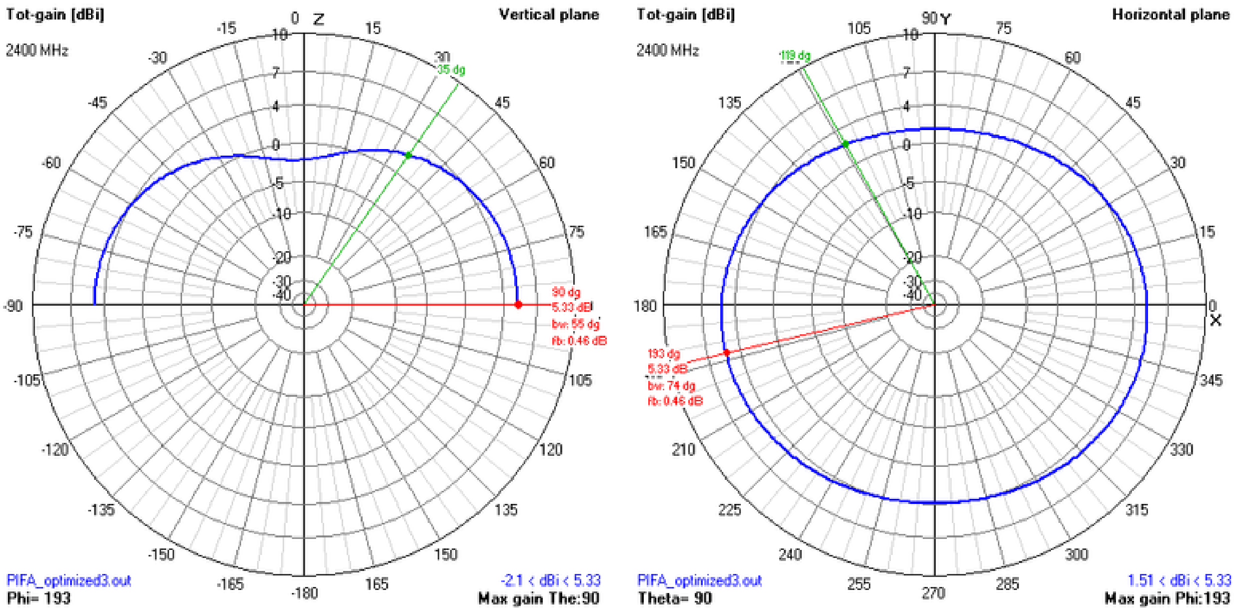


Figure 3: 2D Radiation diagrams

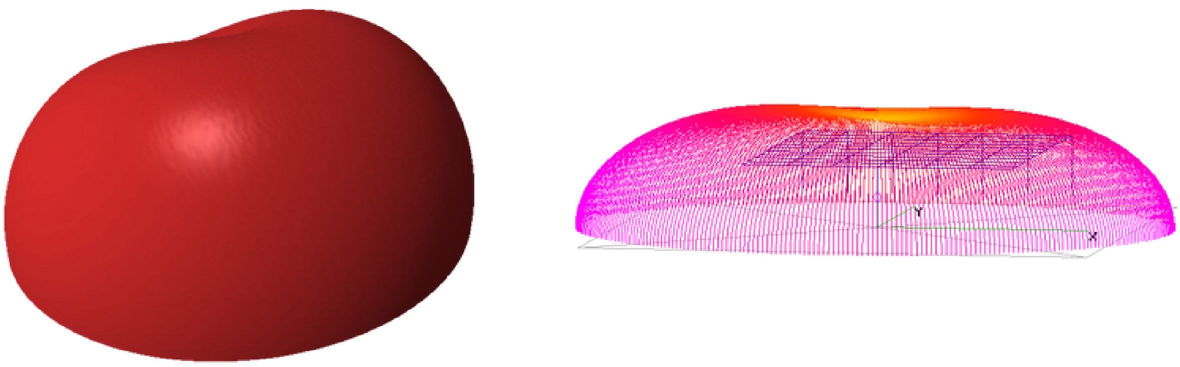


Figure 4: 3D Radiation diagrams

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Questions?
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